

Case Study: Personalized Movie Recommendation System for OTT Platforms

Executive Summary

Modern streaming platforms face a major challenge: helping users discover relevant content among thousands of available movies.

This project developed a Hybrid Movie Recommendation System that leverages machine learning to personalize movie suggestions based on user preferences and historical behavior.

The solution combines Content-Based Filtering and Collaborative Filtering techniques to improve recommendation relevance and user engagement.

Business Problem

Users often experience decision fatigue when browsing large content catalogs.

Common challenges include:

- Difficulty discovering relevant content.
- Reduced user engagement.
- Lower content consumption.
- Increased risk of subscriber churn.

A recommendation system can address these challenges by presenting personalized movie suggestions.

Proposed Solution

A Hybrid Recommendation Engine was developed using:

Content-Based Filtering

Recommendations generated using:

- Movie genres
- Movie tags
- TF-IDF similarity scores

Collaborative Filtering

Recommendations generated using:

- Historical user ratings
- SVD Matrix Factorization

Hybrid Approach

Combined both techniques to provide:

- Personalized recommendations
 - Better recommendation quality
 - Improved user experience
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Implementation Process

Data Preparation

- Dataset cleaning
- Feature engineering
- Tag and genre processing

Model Development

- TF-IDF Vectorization
- Cosine Similarity Matrix
- SVD Model Training

Hybrid Recommendation Layer

Combined content and collaborative scores using weighted ranking.

Evaluation

Metrics used:

- RMSE
- Precision@10
- Recall@10

Results:

- RMSE: 0.8807
 - Precision@10: 0.5660
 - Recall@10: 0.2794
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Key Findings

Personalized Recommendations Improve User Experience

Different users receive different recommendations based on their preferences and behavior.

Hybrid Models Outperform Single Approaches

Combining content-based and collaborative filtering produces more relevant recommendations than using either method independently.

Cold Start Strategy Is Essential

Popularity-based recommendations provide effective recommendations for new users without interaction history.

Business Impact

Potential benefits for an OTT platform include:

Increased Watch Time

Users are more likely to consume content that matches their interests.

Improved Retention

Personalized experiences encourage continued platform usage.

Better Content Discovery

Users can discover relevant content beyond trending titles.

Enhanced Customer Satisfaction

Relevant recommendations improve overall platform experience.

Recommendations

1. Deploy hybrid recommendation systems in production environments.
 2. Continuously collect user feedback and ratings.
 3. Monitor recommendation performance using engagement metrics.
 4. Introduce real-time recommendation updates.
 5. Expand recommendations using user watch history and behavioral signals.
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Conclusion

The Hybrid Movie Recommendation System demonstrates how machine learning can improve content discovery and personalization on streaming platforms. By combining content similarity and user behavior, the solution provides relevant

recommendations that can increase engagement, retention, and customer satisfaction.